Clearance & Elimination Practice 2

Module 4 - Drug Elimination and Clearance · Daily practice sheet · No date

Name _____ Date _____

1a 1. A single 250-mg oral dose of an antibiotic is given to a young man (age 32 years, creatinine clearance 122 mL/min, 78 kg). From the literature, the drug is known to have an apparent V_D equal to 21% of body weight and an elimination half-life of 2 hours. The dose is normally 90% bioavailable. Urinary excretion of the unchanged drug is equal to 70% of the absorbed dose. What is the total body clearance for this drug? (Give it in L/hr.)

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1b 1. A single 250-mg oral dose of an antibiotic is given to a young man (age 32 years, creatinine clearance 122 mL/min, 78 kg). From the literature, the drug is known to have an apparent V_D equal to 21% of body weight and an elimination half-life of 2 hours. The dose is normally 90% bioavailable. Urinary excretion of the unchanged drug is equal to 70% of the absorbed dose. What is the renal clearance for this drug?

1c 1. A single 250-mg oral dose of an antibiotic is given to a young man (age 32 years, creatinine clearance 122 mL/min, 78 kg). From the literature, the drug is known to have an apparent V_D equal to 21% of body weight and an elimination half-life of 2 hours. The dose is normally 90% bioavailable. Urinary excretion of the unchanged drug is equal to 70% of the absorbed dose. What is the probable mechanism for renal clearance of this drug?

A. Filtration only

B. Cleared mainly by the liver

C. Filtration with tubular reabsorption

D. Filtration with active secretion

Problem 2 Estimate the creatinine clearance of a 37-year-old female who weighs 143 lb, serum creatinine of 0.6 mg/dL, and stands 5’6” tall.